

Model Benchmark: Baseline vs Complex Classifiers

Week 7 · Model Optimisation & Trade-offs · Interim Draft

Project	CariSurg MedTech Pathways: assisted ED triage, Mercer General Hospital
Document	Draft model benchmark (Week 7 interim deliverable)
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1 Purpose

This document reports the initial benchmark comparing the Week 6 baseline classifiers against two more complex models on the Mercer emergency-department triage dataset. It is the interim, draft artefact: it presents the measured numbers and the model configurations that produced them. The full cost-benefit analysis, recommendation and risk discussion are delivered separately in the Week 7 cost-benefit memo.

2 Method summary

Dataset: Mercer/Yale ED triage export - 55,121 patient visits, 226 columns. The prediction target is *Triage_Level* (the ESI level, 1–5), where ESI 1 is the sickest and most time-critical.

Class imbalance: The target is severely imbalanced: ESI 1 accounts for just 77 of 55,121 visits (0.14%) approximately 15 patients in the held-out test set. This shapes the interpretation of every ESI-1 figure below.

Leakage control: The raw *esi* field, post-triage outcomes (*disposition*, *previousdispo*) and the row index (*Unnamed: 0*) are removed. All remaining columns, vitals, chief-complaint flags, demographic and administrative fields are retained as inputs, identical to Week 6.

Preprocessing: Text columns are one-hot encoded and numeric columns standardized via a ColumnTransformer inside each model's pipeline.

Evaluation protocol: All models share the same 80/20 stratified split as Week 6 (`random_state = 42`; train 44,096 / test 11,025) and the same preprocessing, so any difference reflects model choice alone. Metrics are reported on the held-out test set; training and inference times are measured in the notebook environment (Google Colab) and are indicative, not production figures.

Primary metric: Recall for ESI 1, the fraction of the sickest patients correctly flagged as justified in Week 6. A missed ESI 1 is the error that most directly harms a patient, so it outranks overall accuracy on this imbalanced task.

Table 1 : Model configurations.

Model	Configuration (scikit-learn)
Dummy (stratified)	DummyClassifier(strategy="stratified", random_state=42)

Model	Configuration (scikit-learn)
Logistic Regression	StandardScaler → LogisticRegression(max_iter=1000, random_state=42)
Decision Tree	DecisionTreeClassifier(max_depth=5, random_state=42)
Random Forest	RandomForestClassifier(n_estimators=300, class_weight="balanced", random_state=42, n_jobs=-1)
Gradient Boosting	HistGradientBoostingClassifier(max_depth=6, learning_rate=0.1, max_iter=300, class_weight="balanced", random_state=42)

3 Six-axis benchmark

Accuracy, macro-averaged precision, recall and F1, the primary ESI-1 recall, and the two cost axes (training time and per-prediction inference time). ★ marks the model recommended in the memo.

Table 2 : Aggregate benchmark across all models (test set, $n = 11,025$).

Model	Accuracy	Precision (macro)	Recall (macro)	F1 (macro)	Recall (ESI 1)	Train (s)	Infer (ms/pred)
Dummy (stratified)	0.3754	0.2041	0.2037	0.2039	0.0000	0.79	0.014
Logistic Regression (baseline) ★	0.7063	0.5882	0.5011	0.5266	0.2500	50.25	0.015
Decision Tree (baseline)	0.5573	0.3533	0.2890	0.2856	0.0000	2.76	0.019
Random Forest	0.6573	0.5129	0.3594	0.3742	0.0000	65.23	0.097
Gradient Boosting	0.5936	0.4279	0.5826	0.4443	0.4375	16.14	0.078

4 Per-class metrics

Aggregate scores mask the rare classes, so precision, recall and F1 are broken out per ESI level. ESI 1 (shaded) is the column that governs the clinical decision.

Table 3 : Recall per ESI level.

Model	ESI 1	ESI 2	ESI 3	ESI 4	ESI 5
Dummy (stratified)	0.0000	0.3275	0.4981	0.1478	0.0453
Logistic Regression (baseline)	0.2500	0.6753	0.7686	0.6594	0.1523
Decision Tree (baseline)	0.0000	0.2510	0.8612	0.3328	0.0000
Random Forest	0.0000	0.6109	0.8238	0.3378	0.0247
Gradient Boosting	0.4375	0.7498	0.4574	0.6965	0.5720

Table 4 : Precision per ESI level.

Model	ESI 1	ESI 2	ESI 3	ESI 4	ESI 5
Dummy (stratified)	0.0000	0.3337	0.4908	0.1471	0.0487
Logistic Regression (baseline)	0.4000	0.7447	0.7071	0.6488	0.4405
Decision Tree (baseline)	0.0000	0.7745	0.5448	0.4471	0.0000
Random Forest	0.0000	0.7242	0.6277	0.6670	0.5455
Gradient Boosting	0.0959	0.6391	0.8005	0.4444	0.1596

Table 5 : F1 per ESI level.

Model	ESI 1	ESI 2	ESI 3	ESI 4	ESI 5
Dummy (stratified)	0.0000	0.3306	0.4944	0.1475	0.0469
Logistic Regression (baseline)	0.3077	0.7083	0.7366	0.6540	0.2263
Decision Tree (baseline)	0.0000	0.3792	0.6674	0.3816	0.0000
Random Forest	0.0000	0.6627	0.7125	0.4485	0.0472
Gradient Boosting	0.1573	0.6900	0.5822	0.5426	0.2496

5 Key observations

- **Logistic regression is the strongest all-rounder** : highest accuracy (0.7063) and macro-F1 (0.5266), and the cheapest inference (~0.015 ms/prediction).
- **The Random Forest does not justify its complexity** : it catches zero ESI-1 patients despite class balancing, trails the baseline on macro-F1 (0.3742), and is the slowest to train (65 s).
- **Gradient Boosting is the only model with real ESI-1 sensitivity** : recall 0.4375 (≈ 7 of ~ 15) and the best macro-recall (0.5826), but its ESI-1 precision is only 0.10, implying roughly ten false criticals per true one.
- **ESI 1 is too rare to evaluate reliably**: with ~ 15 ESI-1 cases in the test set and a single split, all ESI-1 figures carry wide uncertainty and no confidence intervals.
- **Compute cost is not the deciding factor**: every model predicts well under a millisecond per patient, so the choice rests on reliability and interpretability rather than the inference bill.

6 Interpretability

Logistic regression exposes signed coefficients (clear direction of effect per feature); the decision tree is fully traceable but caught no ESI-1 patients; the Random Forest offers only global feature importances (top drivers here: systolic BP, age, diastolic BP, heart rate, glucose); Gradient Boosting is the hardest to narrate case-by-case.

7 Reproducibility & provenance

Source notebook: `notebooks/Week7_Complex_Model_and_Benchmark.ipynb`. Seed 42 throughout. Split: 80/20 stratified on Triage_Level. Companion files:

docs/week7_benchmark_table.md, *docs/week7_perclass_recall.csv*. This is a draft; the final recommendation, arguments and risk register are in the Week 7 cost–benefit memo.